Math 230B Lecture Notes

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1 Week 5

Example ($SL(n,\mathbb{F})$)**.** Suppose $\mathbb{F} = \mathbb{R}$ (or $\mathbb{C}$). We claim that

$$\mathrm{SL}(n,\mathbb{F}) = \{A \in \mathrm{GL}(n,\mathbb{F}) : \det A = 1\}$$

is a Lie subgroup of $\mathrm{GL}(n,\mathbb{F})$.

Indeed, we have that

- $\mathrm{SL}(n,\mathbb{F})$ is closed under multiplication, indeed, if $A, B \in \mathrm{SL}(n,\mathbb{F})$, then

  $$\det(AB) = (\det A)(\det B) = 1 \cdot 1 = 1.$$

  So, AB is invertible and $\det(AB) = 1$. This implies that

  $$AB \in SL(n,\mathbb{F}).$$

- $SL(n,\mathbb{F})$ contains the inverse of each of its elements. Indeed, suppose $A \in SL(n,\mathbb{F})$, then A is invertible and

  $$\det(A^{-1}) = \frac{1}{\det A} = \frac{1}{1} = 1.$$

  So, $A^{-1} \in \mathrm{SL}(n,\mathbb{F})$.

From the above, we can conclude that $\mathrm{SL}(n,\mathbb{F})$ is a subgroup of $\mathrm{GL}(n,\mathbb{F})$.

2 Week 6

Example (Special Orthogonal Group)**.** Note that

$$\begin{aligned}\mathrm{SO}(n) &= \{A \in M(n,\mathbb{R}) : A^T A = I \text{ and } \det A > 0\} \\ &= \{A \in M(n,\mathbb{R}) : A^T A = I \text{ and } \det A = 1\}\end{aligned}$$

We will prove that $\mathrm{SO}(n)$ is a subgroup of $\mathrm{O}(n)$.

- Note that $\mathrm{SO}(n)$ is closed under multiplication. Indeed, suppose A and B are in $\mathrm{SO}(n)$. Our goal is to show that $AB \in \mathrm{SO}(n)$. Then we have

$$A, B \in \mathrm{SO}(n) \implies A, B \in \mathrm{O}(n) \implies AB \in \mathrm{O}(n).$$

Since $A, B \in \mathrm{SO}(n)$, it follows that $\det A = 1$ and $\det B = 1$. Hence,

$$\det AB = (\det A)(\det B) = 1 \cdot 1 = 1.$$

- Also observe that $\mathrm{SO}(n)$ contains the inverse of each of its elements. Indeed, if $A \in \mathrm{SO}(n)$, then

$$A \in \mathrm{SO}(n) \implies A \in \mathrm{O}(n) \implies A^{-1} \in \mathrm{O}(n). \tag{1}$$

Furthermore,

$$A \in \mathrm{SO}(n) \implies \det A = 1 \implies \det A^{-1} = \frac{1}{1} = 1 \implies A^{-1} \in \mathrm{SO}(n) \tag{2}$$

From (1) and (2), we can see that $\mathrm{SO}(n)$ is a subgroup of $\mathrm{O}(n)$. Now, we need to find out whether $\mathrm{SO}(n)$ is a Lie subgroup of $\mathrm{O}(n)$. Note that $\mathrm{SO}(n) = \mathrm{O}(n) \cap \det^{-1}(0, \infty)$ where $(0, \infty)$ is an open set, det is a continuous function, and the preimage of an open set will be open in $M(n, \mathbb{R})$. Hence, $\mathrm{SO}(n)$ will be open in $\mathrm{O}(n)$. Thus, $\mathrm{SO}(n)$ is an embedded submanifold of $\mathrm{O}(n)$ with dimension $\frac{n^2-n}{2}$. Thus, $\mathrm{SO}(n)$ is a Lie subgroup of $\mathrm{O}(n)$ with dimension $\frac{n^2-n}{2}$.

Example (Special Unitary Group)**.** Recall that

$$\mathrm{SU}(n) = \{A \in M(n, \mathbb{C}) : A^*A = I \text{ and } \det A = 1\}.$$

We will show that the above is also a subgroup of $\mathrm{U}(n)$. Clearly, we can see that $\mathrm{SU}(n) \subseteq \mathrm{U}(n)$.

- $\mathrm{SU}(n)$ is closed under multiplication. Indeed, suppose A and B are in $\mathrm{SU}(n)$. Then observe that

$$A, B \in \mathrm{SU}(n) \implies A, B \in \mathrm{U}(n) \implies AB \in \mathrm{U}(n) \tag{1}$$

and

$$\begin{aligned} A, B \in \mathrm{SU}(n) &\implies \det A = 1 \text{ and } \det B = 1 \\ &\implies \det AB = 1 \\ &\implies AB \in \mathrm{SU}(n). \end{aligned}$$

- Also, $\mathrm{SU}(n)$ contains the inverse of each of its elements. Suppose $A \in \mathrm{SU}(n)$. Then

$$A \in \mathrm{SU}(n) \implies A \in \mathrm{U}(n) \implies A^{-1} \in \mathrm{U}(n) \tag{3}$$

and

$$A \in \mathrm{SU}(n) \implies \det A = 1 \implies \det A^{-1} = \frac{1}{\det A} = \frac{1}{1} = 1 \tag{4}$$

Now, (3) and (4) imply that $A^{-1} \in \mathrm{SU}(n)$.

Now, we need to find out if $\mathrm{SU}(n)$ a Lie subgroup of $\mathrm{U}(n)$? Of course! Let $F : \mathrm{U}(n) \to \mathbb{R}$ be defined as follows:

$$F(A) = \varphi.$$

where we previously proved that if $A \in \mathrm{U}(n)$, then $|\det A| = 1$. So, there exists an angle $\varphi \in (-\pi, \pi]$ such that $\det A = e^{i\varphi}$. It can be shown that

$$\forall A \in \mathrm{U}(n) \quad \mathrm{rank} DF(A) = 1.$$

Thus, $\mathrm{SU}(n) = F^{-1}(0)$ is an $n^2 - 1$ dimensional embedded submanifold of $\mathrm{U}(n)$. Therefore, $\mathrm{SU}(n)$ is a Lie subgroup of $\mathrm{U}(n)$ with dimension $n^2 - 1$.

Definition (Bilinear Form). A function $B : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ is said to be a **bilinear form** on $\mathbb{R}^n$ if it satisfies the following condition:

(1) $\forall x, y, z \in \mathbb{R}^n\ B(x + y, z) = B(x, z) + B(y, z)$

(2) $\forall c \in \mathbb{R}$ and $\forall x, y \in \mathbb{R}^n\ B(cx, y) = cB(x, y)$

(3) $\forall x, y, z \in \mathbb{R}^n\ B(x, y + z) = B(x, y) + B(x, z)$

(4) $\forall c \in \mathbb{R}\ \forall x, y \in \mathbb{R}^n\ B(x, cy) = cb(x, y)$

Moreover,

- The bilinear form B is said to be **symmetric** if

$$\forall x, y \in \mathbb{R}^n \quad B(x, y) = B(y, x).$$

- The bilinear form B is said to be **skew-symmetric** if

$$\forall x, y \in \mathbb{R}^n \quad B(x, y) = -B(y, x).$$

- The bilinear form B is said to be **non-degenerate** if the following statement is true

$$B(x, y) = 0 \ \ \forall y \in \mathbb{R}^n \Longrightarrow x = 0.$$

- A symmetric bilinear form B is said to be **positive definite** if

$$B(x, x) > 0 \ \ \forall x \neq 0.$$

Example (Standard Inner Product on $\mathbb{R}^n$). The standard inner product in $\mathbb{R}^n$

$$\langle x, y \rangle = \sum_{i=1}^{n} x_i y_i$$

is a symmetric positive definite bilinear form on $\mathbb{R}^n$.

For example, let's verify condition (1) for bilinearity:

$$\begin{aligned}
\langle x + y, z \rangle &= \sum_{i=1}^{n} [(x_i + y_i) z_i] \\
&= \sum_{i=1}^{n} [x_i z_i + y_i z_i] \\
&= \sum_{i=1}^{n} x_i z_i + \sum_{i=1}^{n} y_i z_i \\
&= \langle x, z \rangle + \langle y, z \rangle.
\end{aligned}$$

I leave you to check the other conditions yourself.

In general, a symmetric positive definite bilinear form on a real vector space V, by definition, is called an **inner product** on V.

Example (Standard Symplectic Form on $\mathbb{R}^n$)**.** Define $w + \mathbb{R}^{2n} \times \mathbb{R}^{2n} \to \mathbb{R}$ as follows:

$$\forall x, y \in \mathbb{R}^{2n} \quad w(x,y) = \sum_{j=1}^{n} x_j y_{n+j} - x_{n+j} y_j.$$

For example,

$$\begin{aligned} n = 1 &\implies w(x,y) = x_1 y_2 + x_2 y_1 \\ n = 2 &\implies w(x,y) = (x_1 y_3 - x_3 y_1) + (x_2 y_4 - x_4 y_2) \\ &\vdots \end{aligned}$$

If we let $\Omega = \begin{pmatrix} O_{n\times n} & I_{n\times n} \\ -I_{n\times n} & O_{n\times n} \end{pmatrix}$, then we can write the standard symplectic form in the following way:

$$w(x,y) = \langle x, \Omega y \rangle$$

where $\langle , \rangle$ is the standard inner product in $\mathbb{R}^{2n}$.

Proposition (Preserving the standard symplectic form on $\mathbb{R}^{2n}$)**.** Suppose $A \in M(2n, \mathbb{R})$. We say that A preserves the standard symplectic form on $\mathbb{R}^{2n}$ if and only if $A^T \Omega A = \Omega$.

Proof. We say that $A \in M(2n, \mathbb{R})$ preserves the standard symplectic form on $\mathbb{R}^{2n}$ if and only if

$$\begin{aligned} \forall x, y \in \mathbb{R}^{2n} \quad w(Ax, Ay) = w(x,y) &\iff \langle Ax, \Omega(Ay) \rangle = \langle x, \Omega y \rangle \\ &\iff \langle x, A^T \Omega A y \rangle = \langle x, \Omega y \rangle \\ &\iff \forall y \in \mathbb{R}^{2n} A^T \Omega A y = \Omega y \\ &\iff A^T \Omega A = \Omega. \end{aligned}$$

■

Proposition. If $A^T \Omega A = \Omega$, then A is invertible.

Proof. Suppose $A^T \Omega A = \Omega$, then we have

$$\begin{aligned} &\det(A^T \Omega)A = \det(A) \\ &\implies \det(A^T)\det(\Omega)\det(A) = \det(\Omega) \\ &\implies (\det(A))^2(\det(\Omega)) = \det(\Omega) \\ &\implies (\det(A))^2 = 1 \\ &\implies \det(A) = \pm 1 \\ &\implies A \text{ is invertible.} \end{aligned}$$

■

Proposition. $A^T \Omega A = \Omega$ if and only if $-\Omega A^T \Omega = A^{-1}$ where A is invertible.

Proof. We have

$$-\Omega A^T\Omega = A^{-1} \underbrace{\iff}_{\text{left multiply by } \Omega^{-1}} -A^T\Omega = \Omega^{-1}A^{-1}$$
$$\underbrace{\iff}_{\text{right multiply by } A} -A^T\Omega A = \Omega^{-1}$$
$$\underbrace{\iff}_{\Omega^{-1}=-\Omega} -A^T\Omega A = \Omega$$
$$\iff A^T\Omega A = \Omega.$$

■

Example (Real Symplectic Group). $\mathrm{Sp}(2n,\mathbb{R}) = \{A \in M(2n,\mathbb{R}) : A^T\Omega A = \Omega\}$ is a subgroup of $\mathrm{GL}(2n,\mathbb{R})$.

Proof. We just proved that every matrix in $\mathrm{SP}(2n,\mathbb{R})$ is invertible. So,

$$\mathrm{SP}(2n,\mathbb{R}) \subseteq \mathrm{GL}(2n,\mathbb{R}).$$

- We prove that $\mathrm{SP}(2n,\mathbb{R})$ is closed under multiplication. Suppose $A, B \in \mathrm{SP}(2n,\mathbb{R})$. We have

$$\begin{aligned}(AB)^T\Omega(AB) &= B^T(A^T\Omega A)B \\ &= B^T\Omega B \\ &= \Omega.\end{aligned}$$

  Hence, we have $AB \in \mathrm{SP}(2n,\mathbb{R})$.

- $\mathrm{SP}(2n,\mathbb{R})$ contains the inverse of each of its elements. Suppose $A \in \mathrm{SP}(2n,\mathbb{R})$. We have

$$\begin{aligned}A^T\Omega A = A &\implies (A^T)^{-1}\Omega A = (A^T)^{-1}\Omega \\ &\implies \Omega = (A^T)^{-1}\Omega A^{-1} \\ &\implies \Omega = (A^{-1})^T\Omega A^{-1} \\ &\implies A^{-1} \in \mathrm{SP}(2n,\mathbb{R})\end{aligned}$$

From the above, we can see that $\mathrm{SP}(2n,\mathbb{R})$ is a subgroup of $\mathrm{GL}(2n,\mathbb{R})$. Now, we will prove that $\mathrm{SP}(2n,\mathbb{R})$ is a Lie subgroup of $\mathrm{GL}(2n,\mathbb{R})$. Define the map

$$G : \mathrm{GL}(2n,\mathbb{R}) \subseteq \mathbb{R}^{4n^2} \to M(2n,\mathbb{R}) \cong \mathbb{R}^{4n^2} \text{ by } A \mapsto A^T\Omega A - \Omega.$$

Clearly, $\mathrm{SP}(2n,\mathbb{R}) = F^{-1}(\{0\})$. It can be shown that

$$\forall A \in \mathrm{GL}(2n,\mathbb{R}) \quad DF(A) \text{ has rank } \frac{4n^2-2n}{2} = 2n^2 - n.$$

Hence, $\mathrm{SP}(2n,\mathbb{R})$ is an embedded submanifold of $\mathrm{GL}(2n,\mathbb{R})$ with dimension $4n^2-(2n^2-n) = 2n^2+n$. Thus, $\mathrm{SP}(2n,\mathbb{R})$ is a Lie subgroup of $\mathrm{GL}(2n,\mathbb{R})$. ■

Definition (Homomorphisms, Isomorphisms, Automorphisms). Let G and H be two groups.

- A function $\varphi : G \to H$ is said to be a **homomorphism** if it preserves group multiplication

$$\forall g_1, g_2 \in G \quad \varphi(g_1g_2) = \varphi(g_1)\varphi(g_2).$$

- A bijective homomorphism is called an **isomorphism**. If there is an Isomorphism between G and G, we say G and H are isomorphic, and we write $G \cong H$.
- An **endomorphism** of G is a homomorphism from G to itself.
- An **automorphism** of G is an isomorphism from G to itself.

Definition (Lie Group Homomorphism). Let G and H be two Lie groups.

- A function $f : G \to H$ is said to be a **Lie group homomorphism** if
  (i) f is a group homomorphism
  (ii) f is smooth
- A function $f : G \to H$ is said to be a **Lie group isomorphism** if
  (i) f is a group isomorphism
  (ii) f is smooth
  (iii) f^{-1} is smooth

  If f is smooth and its inverse is smooth, then f is a diffeomorphism.

Example (The Determinant is a Lie Group Homomorphism). The determinant $\det : \mathrm{GL}(n, \mathbb{R}) \to \mathbb{R}^*$ is a Lie Group Homomorphism:

(i) $\forall A, B \in \mathrm{GL}(n, \mathbb{R})$, we have $\det(AB) = \det(A)\det(B)$. Hence, det is a group homomorphism.

(ii) $\det A$ is a polynomial in terms of the entries of A and so differentiable. Hence, the det is smooth.

Similarly, the $\det GL(n, \mathbb{C}) \to \mathbb{C}^*$ is a Lie Group Homomorphism.

Example (Left Translation/right translation). Let G be a Lie Group and let $g \in G$. The **left translation by** g is the map

$$L_g : G \to G \quad h \mapsto gh.$$

The **right translation by** g is the map

$$R_g : G \to G \quad h \mapsto hg.$$

Notice that L_g is a composition of smooth maps and so it is smooth:

$$G \to G \times G \to G \quad h \mapsto (g, h) \mapsto gh.$$

Starting from left to right, we see that the first mapping is smooth because

- Fact 1: $F = (F_1, F_2)$ is smooth if and only if F_1 and F_2 are smooth.
- Fact 2: $h \mapsto g$ is a constant function implies that it is smooth.
- Fact 3: $h \mapsto h$ is the identity which is also smooth.

The second mapping $G \times G \to G$ is smooth because multiplication in G is smooth since G is a Lie group.

In fact, $L_g : G \to G$ is a diffeomorphism because the inverse L_g is $L_{g^{-1}}$:

$$\forall h \quad L_{g^{-1}} \circ L_g(h) = g^{-1}(gh) = h.$$

Unfortunately, L_g in general is NOT a group homomorphism; that is, in general

$$L_g(h_1h_2) \neq L_g(h_1)L_g(h_2).$$

Similarly, the right translation map also satisfies the properties above but is NOT a group homomorphism.

Example (Conjugation by $g \in G$)**.** Let G be a Lie group and let $g \in G$. The conjugation by g is the map:

$$C_g : G \to G \quad h \mapsto ghg^{-1}.$$

Note that C_g is a diffeomorphism because it is a composition of diffeomorphisms:

$$C_g = L_g \circ R_{g^{-1}}.$$

In fact, $C_g^{-1} = C_{g^{-1}}$. Also, note that

$$\forall h_1, h_2 \in G \quad C_g(h_1 h_2) = g(h_1 h_2)g^{-1} = (gh_1 g^{-1})(gh_2 g^{-1}) = C_g(h_1)C_g(h_2).$$

Hence, $C_g : G \to G$ is a Lie group isomorphism (i.e a Lie group automorphism).

3 Week 7

3.1 Kernel and Image

Definition (Kernel and Image)**.** Let G and H be two groups. Let $f : G \to H$ be a homomorphism.

- The **kernel** of f is the set

$$\ker f = \{g \in G : f(g) = 1_H\} = f^{-1}(\{1_H\}) \subseteq G.$$

- The **image** of f is

$$\mathrm{Im} f = \{f(g) : g \in G\} = f(G) \subseteq H.$$

Both of these sets are subgroups of G and H, respectively.

Example. Consider the determinant $\det : \mathrm{GL}(n, \mathbb{R}) \to \mathbb{R}^*$ is a homomorphism. Thus,

$$\ker(\det) = \{A \in \mathrm{GL}(n, \mathbb{R}) : \det A = 1\} = \mathrm{SL}(n, \mathbb{R}).$$

3.2 A Common Theme in Mathematics

Remark (Topology)**.** If X and Y are two topological spaces, we may consider them to be the same if there exists a function $f : X \to Y$ such that

(1) f is bijective

(2) f is continuous

(3) f^{-1} is continuous

In the context of manifold theory, if M and N are two manifolds, we may consider them to be the same if there exists $f : M \to N$ such that

(1) f is bijective

(2) f is smooth

(3) f^{-1} is smooth

In the context of group theory, if G and H are two groups, we may consider them to be the same if

(1) f is bijective

(2) f is a homomorphism

The last remark gives us an explanation of why (ii) is important.

3.3 Kernel of a Lie Group Homomorphism

Remark (Fact from Linear Algebra)**.** Suppose A and B are two $n \times n$ matrices and A is invertible. Then

$$\operatorname{rank}(AB) = \operatorname{rank}(B) \quad \operatorname{rank}(BA) = \operatorname{rank}(B).$$

Theorem (Kernel of a Lie Group Homomorphism)**.** Let G and H be Lie groups. Let $f : G \to H$ be a Lie group homomorphism. Then

(i) The rank of the derivative of f is constant for all points in G; that is, there exists $k \in \mathbb{N}$ such that for all $x \in G$, $\operatorname{rank}(Df(x)) = k$.

(ii) $\ker(f)$ is an embedded submanifold of G of dimension $\dim(G) - k$ (and so it is a Lie subgroup of G)

(iii) $\operatorname{Im}(f)$ can be equipped with a smooth structure that makes it a Lie subgroup of H.

Proof. Note that (ii) is a direct consequence of (i) and Fact 2 from many weeks ago. Here we will prove (i). It suffices to show that for all $x \in G$, $\operatorname{rank}(Df(x)) = \operatorname{rank}(Df(e))$ where $e = 1_G$. For all $a, x \in G$, we have

$$\begin{aligned} f(x) = f(aa^{-1}x) &= f(a) \cdot f(a^{-1}x) \\ &= (L_{f(a)} \circ f \circ L_{a^{-1}})(x). \end{aligned}$$

By the chain rule,

$$Df(x) = DL_{f(a)}(f(a^{-1}x)) \cdot Df(a^{-1}x) \cdot DL_{a^{-1}}(x).$$

Since $L_{f(a)}$ and $L_{a^{-1}}$ are diffeomorphisms, $DL_{f(a)}$ and $DL_{a^{-1}}$ are invertible at any point. From a fact in linear algebra, for all $x, a \in G$

$$\operatorname{rank}(Df(x)) = \operatorname{rank}(Df(a^{-1}x)).$$

In particular, if we set $a = x$, we get

$$\operatorname{rank}(Df(x)) = \operatorname{rank}(Df(e))$$

which is our desired result. ■

Remark. Suppose G is a diffeomorphism. We have

$$(G^{-1} \circ G)(x) = x \Longrightarrow DG^{-1}(G(x))DG(x) = I$$

Hence, $DG(x)$ is invertible.

3.4 Subgroups and Cosets

Definition (Subgroups and cosets (very important!))**.** Let G be a group. Given a subgroup H of G

and $g \in G$, the **left coset** of H determined by g is the set

$$gH = \{gh : h \in H\}.$$

The **right coset** of H determined by g is the set

$$Hg = \{hg : h \in H\}.$$

Note that H itself is a coset for $g = 1_G$.

You need a subgroup and an element to talk about a coset.

Example ($G = (\mathbb{R}^2, +), H =$ y-axis (Very important!)). Observe that $H = \{(0, y) : y \in \mathbb{R}\}$ is a subgroup of $G = (\mathbb{R}^2, +)$. Let's determine the left cosets of H. For all $(a, b) \in \mathbb{R}^2$, we have

$$\begin{aligned}(a, b) + H &= \{(a, b) + (0, y) : y \in \mathbb{R}\} = \{(a, 0 + y) : y \in \mathbb{R}\} \\ &= \{(a, y) : y \in \mathbb{R}\}.\end{aligned}$$

This represents the vertical line at $x = a$. So, in this example, G decomposes into disjoint cosets. Roughly speaking, the plane is the union of parallel lines. This organizes the mess that is $\mathbb{R}^2$ into more manageable chunks to understand. Is this by accident? NO!

Note that two distinct $g \in G$ can give us the same coset (or vertical line in our example). That is, consider the equivalent cosets

$$(1, 0) + H = (1, 1) + H = \text{ vertical line at } x = a.$$

Theorem. Let G be a group and H be a subgroup of G.

(i) $G = \bigcup_{g \in G} gH$

(ii) Suppose $g_1, g_2 \in G$. Then either $g_1H = g_2H$ or $g_1H \cap g_2H = \emptyset$.

(iii) $g_1H = g_2H$ if and only if $g_1^{-1}g_2 \in H$.

Proof. (i) For all $g \in G$, clearly $gH \subseteq G$ since gH is a subgroup of G. Therefore,

$$\bigcup_{g \in G} gH \subseteq G.$$

Suppose $g' \in G$. Since $1 \in H$, we have $g'1 \in g'H$. Thus, $g' \in g'H \subseteq \bigcup_{g \in G} gH$. So, $G \subseteq \bigcup_{g \in G} gH$. Hence,

$$G = \bigcup_{g \in G} gH.$$

(ii) It suffices to show that if $g_1H \cap g_2H \neq \emptyset$, then $g_1H = g_2H$. Suppose $g \in g_1H \cap g_2H$. Thus,

$$g = g_1h_1 = g_2h_2 \text{ for some } g_1 \text{ and } g_2 \text{ in } H.$$

We have

$$\begin{aligned}g_1 = g_2h_2h_1^{-1} &\implies g_1H = g_2h_2h_1^{-1}H \\ &\implies g_1H = g_2(h_2h_1^{-1}H) \\ &\implies g_1H = g_2H.\end{aligned}$$

(iii) ($\Longleftarrow$) Suppose $g_1^{-1}g_2 \in H$. Our goal is to show that $g_1H = g_2H$. By (ii), it suffices to show that $g_1H \cap g_2H \neq \emptyset$. We have

$$\begin{aligned} g_1^{-1}g_2 \in H &\Longrightarrow g_1(g_1^{-1}g_2) \in g_1H \\ &\Longrightarrow g_2 \in g_1H. \end{aligned}$$

Similarly, we have $g_2 \in g_2H$. Thus, $g_2 \in g_1H \cap g_2H$. So, we have $g_1H \cap g_2H \neq \emptyset$.

($\Longrightarrow$) Assume $g_1H = g_2H$. We claim that $g_1^{-1}g_2 \in H$.

$$\begin{aligned} g_1H = g_2H &\Longrightarrow g_2 \in g_1H \\ &\Longrightarrow g_2 = g_1h \text{ for some } h \in H \\ &\Longrightarrow g_1^{-1}g_2 = h \text{ for some } h \in H \\ &\Longrightarrow g_1^{-1}g_2 \in H. \end{aligned}$$

■

Theorem (Congruence modulo H). Let G be a group and H be a subgroup. Define the relation **congruence modulo** H on G as follows:

$$g_1 \sim g_2 \Longleftrightarrow g_1^{-1}g_2 \in H.$$

Then the above relation is an equivalence relation , and its equivalence classes are precisely the left cosets of H.

Remark.

- When we look at a group G, one very natural perspective is to view it as a disjoint union of cosets of a subgroup H.
- This viewpoint organizes the elements of G and often makes it easier to extract information about the structure of the group.
- In a way, the collection of cosets is simpler and more manageable than the group itself.

Remark.

- It is tempting to try to make this collection of cosets into a group, using the natural operation
$$(g_1H)(g_2H) := (g_1g_2)H.$$
However, this does not always work; in general, the operation may be ill-defined.
- The question is as follows: *Under what conditions is the above operation well-defined?*

Lemma. Let G be a group and H be a subgroup of G. The following statements are equivalent:

(1) $\forall g \in G$, $\forall h \in H$, $ghg^{-1} \in H$

(2) $\forall g \in G$, $\forall h \in H$, $gHg^{-1} = H$

(3) $\forall g \in G$, $gH = Hg$.

3.5 Normal Subgroups

Definition (Normal Subgroups). In the case that any of the above equivalent conditions holds, we say H is a normal subgroup of G.

Notationally, normal subgroups can be represented by $H \trianglelefteq G$. If G is abelian, any subgroup of G is normal.

Theorem. Suppose G is a group and H is a subgroup of G. The operation

$$(g_1H) \cdot (g_2H) = (g_1g_2) \cdot H$$

is well-defined on the collection of left cosets of H if and only if H is a normal subgroup of G.

Proof. ($\Longrightarrow$) Assume that the operation above is well-defined. We want to show that H is a normal subgroup of G. That is, it suffices to show that for all $g \in G$ and $h \in H$, we have

$$ghg^{-1} \in H.$$

That is, we want to show that (using the previous result $aH = bH \iff b^{-1}a \in H$)

$$g^{-1}H = (gh)^{-1}H.$$

That is, we want to show that

$$g^{-1}H = h^{-1}g^{-1}H \tag{*}$$

Since the coset operation is well defined, we have

$$\begin{aligned} g^{-1}H = (1g^{-1})H = (1H)(g^{-1}H) &\underbrace{=}_{1H=H=h^{-1}H} (h^{-1}H)(g^{-1}H) \\ &= (h^{-1}g^{-1})H. \end{aligned}$$

($\Longleftarrow$) Assume that $H \trianglelefteq G$. We want to show that the above operation is well-defined. Suppose $g_1H = g_1'H$ and $g_2H = g_2'H$. Our goal is to show that

$$(g_1g_2)H = (g_1'g_2')H. \tag{*}$$

Since H is normal, (that is, $g_2'H = Hg_2'$ and $g_1'H = Hg_1'$)

$$\begin{aligned} g_1'g_2'H = g_1'Hg_2' &= g_1Hg_2' && (g_1'H = g_1H) \\ &= g_1g_2'H && (H \trianglelefteq G \Longrightarrow Hg_2' = g_2'H) \\ &= g_1g_2H. && (g_2'H = g_2H) \end{aligned}$$

and we are done. ■

Theorem (Quotient Theorem for Groups)**.** If H is a normal subgroup of G, then the operation

$$(g_1H) \cdot (g_2H) = (g_1h_2)H$$

is well-defined on cosets

The group of cosets is called the **quotient group of G by H**, and is written G/H.

Remark. The map $\pi : G \to G/H$ where $g \mapsto gH$ is a surjective homomorphism and is called the **quotient map**.

By studying quotient groups of a group which, in a sense, organizes the group's elements into simpler chunks, we can gain valuable information about the group itself. For example, we can show that if $G/Z(G)$ is cyclic, then G is abelian (where $Z(G)$ is the center of G).

The kernel of the quotient map will be a normal subgroup of G.

3.6 Making G/N a Lie Group

Remark (Observation). Suppose G is a Lie group and N is a normal algebraic subgroup of G. From this we can construct a new algebraic group G/N. In the context of Lie Theory, an important question arises:

Can we make G/N a Lie group?

It seems reasonable to look for a smooth structure on G/N that at least ensures that the quotient map $\pi : G \to G/N$ is smooth.

What is the kernel of π? It is

$$\begin{aligned}\ker(\pi) = \pi^{-1}\{1_G N\} &= \{g \in G : \pi(g) = 1_G N\}\\ &= \{g \in G : \pi(g) = N\}\\ &= \{g \in G : gN = N\}\\ &= N.\end{aligned}$$

(**very important!**)

What can we conclude from this result? If we manage to find a smooth structure on G/N that ensures $\pi : G \to G/N$ is smooth, then N must be necessarily a topologically closed subset of G (recall that if π is smooth, then it is continuous and preimages of continuous functions of closed sets are closed). If it is not closed, then there is no way to define a smooth structure on the homomorphism π.

Theorem (Quotient Theorem for Lie Groups). Let G be a Lie group, $N \trianglelefteq G$, and N is topologically closed subset of G. Then G/N can be equipped with a smooth structure that turns it into a Lie group such that the quotient map $\pi : G \to G/N$ is a surjective smooth map.

Example ($\mathbb{R}/\mathbb{Z}$). Note that $G = (\mathbb{R}, +)$ is a Lie group and $N = (\mathbb{Z}, +)$ is a subgroup of G. Since G is abelian, $N \trianglelefteq G$. Now, is N a closed subset? Yes, because the complement of $\mathbb{Z}$ (that is $\mathbb{R} \setminus \mathbb{Z}$) is open, so $\mathbb{Z}$ is a closed subset of $\mathbb{R}$. Using the Quotient theorem for Lie groups, we have that $\mathbb{R} \setminus \mathbb{Z}$ is a Lie group.

As a matter of fact, the mapping

$$\begin{aligned}F : \mathbb{R}/\mathbb{Z} &\longrightarrow S^1 = \mathrm{U}(1)\\ x + \mathbb{Z} &\longmapsto e^{2\pi i x}\end{aligned}$$

is a well-defined isomorphism of Lie group. And so,

$$\mathbb{R}/\mathbb{Z} \cong S^1.$$

Why is this well-defined? Suppose $x + \mathbb{Z} = y + \mathbb{Z}$. We need to show that $e^{2\pi i x} = e^{2\pi i y}$. We have

$$\begin{aligned}x + \mathbb{Z} = y + \mathbb{Z} &\Longrightarrow x - y \in \mathbb{Z}\\ &\Longrightarrow e^{2\pi i(x-y)} = \cos(2\pi(x-y)) + i\sin(2\pi(x-y))\\ &\Longrightarrow e^{2\pi i(x-y)} = 1 + i0 = 1\\ &\Longrightarrow e^{2\pi i x} \cdot e^{-2\pi i y} = 1\\ &\Longrightarrow e^{2\pi i x} = e^{2\pi i y}.\end{aligned}$$

Similarly, we can show that

$$\mathbb{R}^n/\mathbb{Z}^n \cong (S^1) \times \cdots \times (S^1) \qquad (n\text{-dimensional torus})$$

mapped by $(x_1, \ldots, x_n) + \mathbb{Z}^n \mapsto (e^{2\pi i x_1}, e^{2\pi i x_2}, \ldots, e^{2\pi i x_n})$.

Example. Let $G = \mathrm{SU}(2)$ be a Lie group, $N = \{-I, +I\}$ is a normal subgroup of $\mathrm{SU}(2)$, then

$$\forall g \in \mathrm{SU}(2) \quad g(\pm I)g^{-1} = \pm gg^{-1} = \pm I \in N$$

where N is closed G (N is finite subset of G). Hence, $\mathrm{SU}(2)/\{-I, I\}$ is a Lie group. In fact,

$$\mathrm{SU}(2)/\{-I, +I\} \cong \mathrm{SO}(3).$$